

SpringAIS: Consulting Meeting Brief

Senior Partner Review - Valent

Date: 2025-12-20

Prepared for: Senior Partner, Valent

Purpose: Project review and strategic feedback

Project: SpringAIS - AI-Driven Talent Mobility Platform

Executive Summary

SpringAIS is an AI-powered internal talent mobility platform designed for EY's enterprise talent optimization challenge. We're building this for the **EY Artificial Intelligence Competition** at SCLC 2026 (submission deadline: February 16, 2026). The platform uses semantic AI to match employees to roles beyond keyword matching, generates personalized upskilling paths, and provides explainable recommendations with bias mitigation and privacy safeguards.

Core Innovation: Unlike traditional HR systems that match skills to job descriptions, SpringAIS reveals what actually drives career advancement—combining skills, performance metrics, behavioral patterns, and success benchmarks from employees who successfully advanced.

Competition Context

Source: EY Artificial Intelligence Competition - SCLC 2026

Problem Statement

Enterprises need agile talent strategies. Traditional HR systems fail to: - Map evolving skills to open roles - Provide actionable upskilling paths - Balance AI automation with governance (bias, explainability, privacy)

Competition Requirements

1. **Working Prototype:** Skill-role matching + upskilling plans for 5+ synthetic profiles (LLM-based approach required)
2. **Explainability & Governance:** Decision logs, bias checks, privacy safeguards
3. **Presentation:** ≤10-slide deck + demo (video or live)

Evaluation Rubric (100 points)

- **AI Functionality & Accuracy (20 pts):** Accurate skill-role mapping beyond keyword matching
- **Explainability & Governance (20 pts):** Transparent recommendations, bias detection, privacy safeguards
- **Technical Design (20 pts):** Well-structured, scalable, addresses IT security and AI risks
- **Problem Understanding & Business Value (15 pts):** Real enterprise need, meaningful value
- **User Experience & Presentation (15 pts):** Clear demo, engaging storytelling
- **Innovation & Creativity (10 pts):** Unique, forward-looking approach

Timeline: - February 16, 2026 - Preliminary submissions due - February 26, 2026 - Finalists notified - March 27, 2026 - Final presentations & winners announced

Prizes: \$2,000 (1st), \$1,000 (2nd), \$500 (3rd)

EY Structure Analysis

Career Progression Model

Standard Hierarchy: Staff → Senior → Manager → Senior Manager → Partner/Executive Director

Progression Timelines (varies by business unit): - Consulting: 2 → 2-3 → 2-4 → 4-8 years - Tax: 2-3 → 2-3 → 3-4 → 6-8 years - Audit: 3 → 2-3 → 3 → 2-5+ years

EY-Parthenon Exception: Different titles (Associate → Senior Associate → Consultant → Director → Senior Director → Partner)

Promotion Cycles

- **Regular Promotions:** August (aligned with fiscal year end, July-June)
- **Agile Promotions:** January (rank changes only)
- **Calibration Sessions:** Late May/June (decisions made ~3 months before effective date)

Success Factors (Beyond Skills)

Six Metric Categories: 1. **Financial:** Effective utilization (decreases with seniority: 95% → 70%), billable hours, realization rate 2. **Compliance:** Timesheet compliance (95%+), CPE hours (40+), policy adherence 3. **Quality:** Engagement ratings, technical excellence 4. **Development:** Learning hours, mentoring participation, EY Badges (87 badges, 5 tiers: Learning → Bronze → Silver → Gold → Platinum) 5. **People:** Upward feedback, team scores, mentee count 6. **Feedback Themes:** NLP analysis of recurring themes (leadership, client management, technical depth)

Critical Insight: Utilization targets *decrease* as seniority increases (95% Staff → 70% Partner), reflecting shift from billable work to business development and strategic activities.

Soft Factors: - **Sponsor/Advocate:** Someone who fights for you in calibration sessions (politics matter) -

Visibility Moves: Internal community leadership, thought leadership, mentoring - **Personal Brand:** “Go-To Expert” specialization accelerates advancement

Internal Mobility Context

- **Mobility4U Program:** ~900 employees started new assignments, 4,100+ on mobility assignments
- **Fear of Discovery:** Employees hesitate to explore internal opportunities (manager might find out)
- **Service Line Translation:** Skills translate across service lines (e.g., Audit → Tech Risk, Tax → Advisory)

EY Technology Stack

- **SuccessFactors:** Core HR platform (employee profiles, performance, learning)
- **EY PX360:** Experience data (X-data) + Operational data (O-data) integration
- **Credly:** Digital badge verification (87 badges, OAuth 2.0 API)

- **LEAD Framework:** Performance management system (launched 2018)
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Proposed Solution: SpringAIS

What We're Building

SpringAIS is an AI-powered career discovery and development platform that: 1. **Reveals hidden opportunities** employees didn't know existed (semantic matching across all service lines) 2. **Shows exactly how to get there** with actionable, time-estimated upskilling paths 3. **Provides motivation and clarity** by comparing employees to success patterns from those who advanced

How It Works

Phase 1: Discovery - Employees upload resume, Credly badges, project descriptions - **Dual LLM validation** extracts skills WITH evidence quotes (LLM #1 extracts, LLM #2 validates) - **Pure vector semantic matching** finds role alignments using GPT-5.2 embeddings (1536 dimensions) - Discovery modes: Best Fit (70%+), Stretch (50-70%), Exploratory (unexpected pivots), Trending (high-demand areas) - **Anonymous exploration:** Employees explore without manager visibility

Phase 2: Career Journey Map - Interactive skill tree visualization (React Flow) - **Success Pattern Overlay:** "Employees who advanced to Manager typically showed: 87% effective utilization (you: 78%), 2+ mentees (you: 0), feedback themes emphasizing leadership..." - **Career Competitiveness Dashboard:** Visual indicators across 6 metric categories - **Nine Box Position:** Performance × Potential matrix

Phase 3: Actionable Development - Personalized upskilling paths with time estimates (e.g., "AWS cert: 3-4 months, 120 study hours") - Progress visualization: "50% match → 70% if you complete X, Y, Z" - Holistic recommendations: skills + behaviors + visibility moves

Two-Sided Anonymous Matching: - Hiring manager posts role → System shows candidate COUNT (not identities) - Employees opt-in → Manager sees tokenized profiles (EMP-482910) - Identity revealed only after manager invites conversation

Why This Approach

1. Semantic AI, Not Keyword Matching - Traditional systems break on synonyms ("cloud architecture" vs "AWS/Azure") - Vector embeddings understand skill relationships automatically - No manual skill normalization required

2. Dual LLM Validation for Explainability - Addresses AI hallucination concerns - Every inferred skill shows supporting evidence quote + confidence level - Human-readable explanations: "Inferred Python expertise because resume states: [quote]"

3. Success Pattern Analysis (The Breakthrough) - Competitors stop at skill matching - We analyze what ACTUALLY drives advancement (6 metric categories) - Shows employees where they are vs. where successful employees were - Transforms vague feedback ("need more visibility") into concrete actions

4. Privacy-First Architecture - Tokenization (EMP-482910) replaces PII throughout matching - Anonymous exploration until mutual opt-in - Audit trails for compliance

5. Legally Defensible, Ethically Sound - "Patterns not promises" language throughout - Confidence intervals, not absolute predictions - Reason codes for every recommendation - Bias monitoring and disparate

impact testing

Technical Architecture

Technology Stack

Backend: - **FastAPI (Python):** High-performance async REST API - **GPT-5.2 Instant:** Skill inference, validation, embeddings (400K context window) - **LangChain:** LLM orchestration, aggressive caching (semantic + prompt caching = 68.8% API call reduction) - **PostgreSQL + pgvector:** Unified structured + vector data storage - **Chroma (demo) / Qdrant (production):** Vector database for semantic search - **Redis:** Multi-layer caching (LLM responses, embeddings, match results)

Frontend: - **React + TypeScript:** Component-based UI with type safety - **shadcn/ui:** Professional UI component library - **React Flow:** Interactive skill tree visualization - **Recharts:** Analytics dashboards

Infrastructure: - **Docker + docker-compose:** Single-command deployment (docker-compose up) - **Container Architecture:** Backend, Frontend, PostgreSQL, Chroma, Redis

Key Technical Innovations

1. Dual LLM Validation Pattern - LLM #1: Extract skills WITH evidence quotes from source documents - LLM #2: Independently validate that quote actually supports inferred skill - Output: Confidence scores (high/medium/low) for every skill, with human-readable evidence - **Why:** Eliminates hallucinations, builds trust through explainability

2. Pure Vector Semantic Matching - GPT-5.2 embeddings (1536-dimensional semantic space) - Skills that are semantically related cluster together in vector space - Chroma vector database for local, reliable matching - **Why:** Handles synonyms, skill hierarchies, related competencies automatically

3. Multi-Layer Aggressive Caching - **Semantic Cache:** Similar query embeddings → cached responses (68.8% API call reduction) - **Prompt Cache:** Repeated prompt prefixes >1024 tokens (90% cost reduction) - **Response Cache:** Exact skill inference results (7 days TTL) - **Embedding Cache:** Generated embeddings per skill/document (indefinite) - **Why:** Manages LLM API costs, improves response times

4. Hybrid Data Architecture - PostgreSQL + pgvector for structured data + embeddings - Optional dedicated vector DB (Chroma/Qdrant) for advanced semantic search - Redis for caching layer - **Why:** Flexibility to start simple, scale to production

Integration Architecture (MVP: Mock Data)

SuccessFactors Mock: - OData V4-compatible JSON structure - Employee profiles, performance metrics, learning records - Fiscal year alignment (July-June)

Credly Mock: - OAuth 2.0-style badge metadata - 87 badges, 5-tier structure (Learning → Bronze → Silver → Gold → Platinum) - Skill tags and issue dates

Future-Ready Design: - Mock data served through same API interface as production - Configuration flag to switch between mock and live data sources - Data adapters abstract the source (mock vs. real API)

Performance Benchmarks

- **Chroma vector queries:** <350ms p95 (demo scale)
 - **Cached skill inference:** <3s (semantic cache hit)
 - **Uncached skill inference:** <15s (full dual LLM pipeline)
 - **Role matching:** <2s for top-10 results
 - **Total memory footprint:** ~4.5GB for full stack
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Key Differentiators

What Makes SpringAIS Special

1. **Success Pattern Analysis:** No competitor captures what actually drives advancement (6 metric categories, behavioral patterns, soft factors)
 2. **Dual LLM Validation:** Explainable AI with evidence quotes—employees can see WHY skills were inferred
 3. **Pure Vector Semantic Matching:** Handles skill relationships automatically (no manual synonym lists)
 4. **Privacy-First Anonymous Matching:** Employees explore safely without manager visibility until ready
 5. **Career Journey Map Visualization:** Transforms abstract career advice into concrete, motivating progression paths
 6. **EY-Specific Deep Integration:** Aligned with EY's promotion cycles, calibration processes, badge system, service line structures
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Questions for Feedback

Strategic Questions

1. **Business Value:** Does this solve a real enterprise need? Is the value proposition clear and compelling?
2. **Competition Positioning:** How does this compare to typical AI/HR solutions? Is the differentiation strong enough?
3. **EY Context:** Are we accurately representing EY's structure and processes? Any critical gaps in our understanding?
4. **Success Pattern Approach:** Is analyzing what actually drives advancement (beyond skills) a compelling differentiator?

Technical Questions

5. **Architecture:** Is the technical approach sound? Any concerns about scalability, security, or AI risks?
6. **Dual LLM Validation:** Is the explainability approach sufficient? Does it address hallucination concerns effectively?

7. **Caching Strategy:** Is the multi-layer caching approach appropriate for managing LLM API costs?

Competition Readiness

8. **Demo Strategy:** What should we emphasize in the ≤10-slide presentation? What’s the most compelling “holy shit” moment?
9. **Governance & Explainability:** Are our bias mitigation and privacy safeguards sufficient for the competition rubric?
10. **Timeline:** Given the February 16 deadline, are we prioritizing the right features? What’s critical vs. nice-to-have?

Risk Assessment

11. **Technical Risks:** What are the biggest technical risks we should mitigate? (LLM hallucinations, vector matching quality, performance)
12. **Competition Risks:** What could go wrong in the demo? How should we prepare for edge cases or judge questions?

Appendix: Competition Rubric Alignment

Rubric Category	How SpringAIS Addresses It
AI Functionality & Accuracy (20 pts)	Dual LLM validation ensures accuracy; pure vector semantic matching goes beyond keyword matching; success pattern analysis provides meaningful recommendations
Explainability & Governance (20 pts)	Evidence quotes for every skill inference; reason codes for all matches; bias detection framework; privacy safeguards (tokenization, audit logs); “patterns not promises” language
Technical Design (20 pts)	Well-structured monolithic architecture (microservices-ready); comprehensive documentation; addresses IT security (HTTPS, encryption, RBAC) and AI risks (bias mitigation, hallucination prevention)
Problem Understanding & Business Value (15 pts)	Deep EY structure analysis; addresses real enterprise need (internal mobility, retention, cost reduction); clear value proposition (10-20% internal fill rate lift, 30-50% time-to-fill reduction)
User Experience & Presentation (15 pts)	Professional UI (shadcn/ui); clear user journeys; engaging Career Journey Map visualization; strong storytelling (Maya’s journey from invisible progress to promotion clarity)
Innovation & Creativity (10 pts)	Success pattern analysis (unique approach); dual LLM validation (explainable AI innovation); pure vector semantic matching (beyond keyword matching); privacy-first anonymous matching

Document Prepared By: SpringAIS Team

Next Steps: Incorporate partner feedback, refine solution, prepare competition submission